Field image calibration workflow – fieldwork guide

Last updated by Shunan on 2026-07-08

This guide describes a standardized workflow for capturing surface ice imagery with consumer digital cameras and calibrating color using a physical color checker. The goal is to obtain color consistent images suitable for post-processing and comparison across sites, dates, and users.

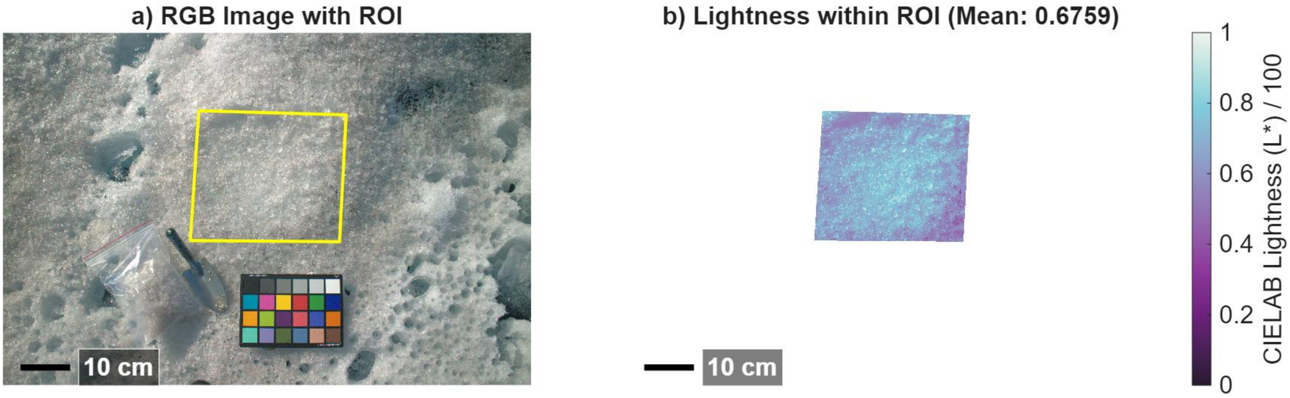

Figure 1 Example of postprocessing of field images

Required:

1) Digital camera with RAW capability
   a. Must support true RAW formats (e.g., .CR2, .NEF, .DNG, etc.)
   b. JPEG-only or HEIF formats (e.g., standard iPhone images) are NOT acceptable as RAW
2) A color checker (e.g., 24-color Calibrite ColorChecker)

Camera Setup (Before Fieldwork)

1) Use consistent settings throughout the campaign whenever possible.
2) File format: RAW and JPEG (JPEG optional for reference only but good to have)
3) White balance: Fixed (e.g., Direct sunlight or Custom). Do not use Auto White Balance
4) Exposure mode: Manual exposure (preferred)
5) Flash: Off
6) Focus: Manual or single-shot autofocus and ensure ice surface and color checker are sharp

In the field

Prefer uniform lighting conditions (overcast or high sun). Avoid low sun angles, mixed sun–cloud conditions. Record general sky conditions (clear, thin cloud, overcast)

7) Place the colorchecker flat (if the terrain allows) on the ice surface and ensure it is fully illuminated. Avoid shadows, reflections, snow contamination, or nearby obstacles.
8) Camera positioning (recommended if possible):

   a. Hold the camera in a nadir (downward-looking) position if possible. Sometimes we have to alter the angle a bit to avoid shading .
   b. Keep the sensor plane as parallel to the surface as possible.
   c. Maintain a consistent height above the surface.

9) Exposure adjustment
   a. If possible, adjust exposure to avoid saturation, especially in the white patches of the color checker and bright ice (if it's too bright).
   b. Check the histogram (preferably centered) and ensure no channel is saturated, if your camera supports this.

10) Metering (you will see it in the camera setting if your camera supports this):
   a. Use center-weighted or spot metering on the ice surface near the color checker.
   b. Avoid evaluative/matrix metering in bright snow or ice scenes.

11) Capture the image
   a. Ensure the entire color checker is visible in the frame.
   b. Keep the color checker and sampling area under similar illumination conditions.
   c. Take at least one well-exposed RAW image per site.
   d. It is strongly recommended to take another photo after sampling to make it easier to select your ROI in the postprocessing.

12) Additional images are recommended if illumination conditions change.

Done!

A desktop application was developed to postprocess RAW images on Windows/macOS/Linux. It is available on GitHub: https://github.com/glacier-lab/Field-Image-Calibration.